Endpoint maximality of a three-point Riesz energy on the circle: a proof of Fan's Conjecture 5.9

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28 October 2026

Abstract

Fix $r \geq 2$ and $\pi/2 < \varphi \leq 2\pi/3$. Let $B = e^{i\varphi}$ and $C = e^{-i\varphi}$ on the unit circle, and let $A = e^{i\psi}$ with $0 \leq \psi \leq 2\pi - 3\varphi$. Write $E_r(\psi;\varphi)$ for the largest value of $\sum_{i,j} m_i m_j |x_i - x_j|^r$ over probability measures on $\{A, B, C\}$. Q. Fan conjectured that this energy, equivalently the Riesz $(-r)$-capacity, is maximized at the endpoint $\psi = 2\pi - 3\varphi$, where $|A-C| = |B-C|$ (Conjecture 5.9 of *Riesz capacity ratios with negative exponents*, arXiv:2609.11186v1). We prove the conjecture on its whole domain. For $r = 2$ the energy is identically 2. For $r > 2$ the endpoint is the unique maximizer, and the maximum is $2^{r-1}\sin^r\varphi/(1 - 2^{r-2}|\cos\varphi|^r)$. The proof does not use monotonicity in ψ. After an exact solution of the three-weight problem, both branches of the optimized energy compare with the endpoint through one algebraic identity. The comparison then reduces to the two-term majorization inequality $p^t + q^t \leq k^t + s^t$, where $t = r/2$ and $k + s = p + q$, applied to the squared half-chords of the triangle. When $0 < r < 2$ the same identity shows that the endpoint is instead the unique minimizer. Combined with Björck's theorem that, for $r > 2$, planar equilibrium measures may be taken on at most three points, the result also proves Fan's Conjecture 5.6. That conjecture gives the energy and an equilibrium measure of the vertex set of a regular odd-gon.

1 Introduction

For a compact set $K \subset \mathbb{R}^n$ and $r > 0$, the Riesz energy with negative exponent $p = -r$ is

$$U_r(K) = \max_{\mu} \iint_{K\times K} |x - y|^r \, d\mu(x)\, d\mu(y),$$

the maximum being taken over Borel probability measures on K. The Riesz $(-r)$-capacity is $\mathrm{Cap}_{-r}(K) = U_r(K)^{1/r}$ [4, 2], and a maximizing measure is an equilibrium measure. Since $u \mapsto u^{1/r}$ is strictly increasing, every comparison of capacities in this paper is the same as a comparison of energies. For a finite set $\{x_1, \ldots, x_m\}$ the energy is the maximum of the quadratic form $\sum_{i,j} m_i m_j |x_i - x_j|^r$ over the probability simplex.

Clark and Laugesen [2] developed a program for maximizing capacity ratios. Fan [4] studied negative exponents. One question there is which vertex triple carries an equilibrium measure of the vertex set P_N of a regular N-gon when N is odd. Fan reduced it to two steps. The first is a symmetric comparison of isosceles triangles, which Fan proved [4, Lemma 5.8]. The second is the following statement, which Fan left open [4, Section 5.4].

Conjecture 1.1 (Fan [4, Conjecture 5.9])**.** *Suppose $r \geq 2$. Let $B = (\cos\varphi, \sin\varphi)$ and $C = (\cos\varphi, -\sin\varphi)$ with $\varphi \in (\pi/2, 2\pi/3]$. Let $A = (\cos\psi, \sin\psi)$ with $\psi \in [0, 2\pi - 3\varphi]$, so that BC remains a longest side of the triangle. Then the capacity of $\{A, B, C\}$ is maximized when $\psi = 2\pi - 3\varphi$, that is, when $|A - C| = |B - C|$.*

Writing u, v, w for the masses at A, B, C, the energy of the three-point set is

$$E_r(\psi;\varphi) = \max_{\substack{u,v,w\geq 0\\ u+v+w=1}} 2(uv|A-B|^r + uw|A-C|^r + vw|B-C|^r). \tag{1}$$

Throughout, $\psi_{\mathrm{e}} = \psi_{\mathrm{e}}(\varphi) := 2\pi - 3\varphi$ and $A_{\mathrm{e}} = e^{i\psi_{\mathrm{e}}}$.

Theorem 1.2 (Main theorem)**.** *Let $r \geq 2$, $\pi/2 < \varphi \leq 2\pi/3$ and $0 \leq \psi \leq \psi_{\mathrm{e}}$. Then*

$$E_r(\psi;\varphi) \leq E_r(\psi_{\mathrm{e}};\varphi) = \frac{2^{r-1}\sin^r\varphi}{1-2^{r-2}|\cos\varphi|^r}. \tag{2}$$

Moreover:

(i) *if $r = 2$, then $E_2(\psi;\varphi) = 2$ for every admissible (φ,ψ), so every point of the arc is a maximizer;*

(ii) *if $r > 2$ and $\psi < \psi_{\mathrm{e}}$, the inequality (2) is strict, so $\psi = \psi_{\mathrm{e}}$ is the unique maximizer on $[0,\psi_{\mathrm{e}}]$;*

(iii) *at $\psi = \psi_{\mathrm{e}}$ the maximum in (1) is attained by the masses $u = v = c/(4c-e)$ at A, B and $w = (2c-e)/(4c-e)$ at C. Here $c = |B-C|^r$ and $e = |A_{\mathrm{e}} - B|^r$, and all three masses are positive.*

In particular Conjecture 1.1 holds for every admissible (r,φ,ψ).

The theorem asserts endpoint maximality. It does not assert that $E_r(\cdot\,;\varphi)$ is monotone in ψ, and we make no claim about monotonicity. Fan's numerical observations [4, Section 5.4] suggest that the energy is constant and then increasing. When $\varphi = 2\pi/3$ the interval $[0,\psi_{\mathrm{e}}]$ reduces to the single point $\psi = 0$ and the triangle is equilateral. The statement is then trivial, but every formula below remains valid there.

The idea. Fan's attempt tracked where the optimal measure switches between two and three points. That switch quantity is not monotone in ψ [4, Section 5.4.1]. We avoid the switch entirely. Put $p = \sin^2\frac{\varphi-\psi}{2}$, $q = \sin^2\frac{\varphi+\psi}{2}$, $s = \sin^2\varphi$ (squared half-chords) and $k = p + q - s$. The law of cosines on the circle gives $k > 0$, $k^2 = 4\cos^2\varphi\, pq$ and $k \leq p \leq q \leq s$. An exact identity reduces the comparison with the endpoint to

$$p^t + q^t \leq s^t + k^t, \qquad t = r/2 \geq 1.$$

Since (k,s) majorizes (p,q), this is just convexity of $u \mapsto u^t$. For $r = 2$ it is an equality, the law of cosines. For $0 < r < 2$ the inequality reverses (Proposition 3.4).

Consequences. Section 4 restates Theorem 1.2 for an arbitrary triangle inscribed in the unit circle. It then deduces Fan's odd-polygon Conjecture 5.6, using Björck's support theorem as cited in [2, 4]. Fan already indicated this deduction [4, after Conjecture 5.9]; we give the details.

2 Maximizing over three weights

Let $a, b, c > 0$. With masses u, v, w at three points whose pairwise kernel values are a (between the points of u and v), b (between u and w) and c (between v and w), set

$$Q(u,v,w) = 2(uva + uwb + vwc) = x^{\mathsf{T}}Mx, \qquad M = \begin{pmatrix} 0 & a & b \\ a & 0 & c \\ b & c & 0 \end{pmatrix}, \quad x = (u,v,w)^{\mathsf{T}},$$

and let $E = \max_\Delta Q$ over the closed simplex $\Delta = \{x \geq 0 : u + v + w = 1\}$. Put

$$D = 2ab + 2bc + 2ca - a^2 - b^2 - c^2 = 4ab - (a + b - c)^2, \qquad F = \frac{2abc}{D} \quad (D \neq 0). \tag{3}$$

Lemma 2.1. *Either $E = \max\{a/2, b/2, c/2\}$, or $E = F$ and the maximum is attained at the point*

$$x^* = \frac{1}{D}\big(c(a + b - c),\ b(a + c - b),\ a(b + c - a)\big), \tag{4}$$

which then lies in the open simplex. In all cases $E \geq \max\{a/2, b/2, c/2\}$.

Proof. On the edge $w = 0$ we have $Q = 2uva \leq a(u+v)^2/2 = a/2$, with equality at $u = v = \frac{1}{2}$; the other edges are similar, and $Q = 0$ at the vertices. Hence $E \geq \max\{a, b, c\}/2$. The simplex is compact, so Q attains its maximum at some $\bar{x}$. If $\bar{x}$ lies on the boundary, then $E \leq \max\{a, b, c\}/2$ by the edge bound, so E equals this number. Otherwise $\bar{x}$ is an interior local maximum of Q on the plane $u + v + w = 1$. By the Lagrange condition, $\nabla Q = 2M\bar{x}$ is parallel to $(1, 1, 1)$, that is, $M\bar{x} = \lambda\mathbf{1}$ for some λ. Now $\det M = 2abc \neq 0$, so $\bar{x} = \lambda M^{-1}\mathbf{1}$. A direct computation gives $\mathbf{1}^{\mathsf{T}} M^{-1}\mathbf{1} = D/(2abc)$. Since $\mathbf{1}^{\mathsf{T}}\bar{x} = 1$, we get $D \neq 0$, $\lambda = 2abc/D = F$ and $\bar{x} = x^*$; one checks directly that (4) satisfies $Mx^* = F\mathbf{1}$ and has coordinate sum 1. Finally $E = Q(\bar{x}) = \bar{x}^{\mathsf{T}} M\bar{x} = \lambda\,\mathbf{1}^{\mathsf{T}}\bar{x} = F$. □

The next lemma recovers the three-point formula of Clark and Laugesen [2, Theorem 8], quoted by Fan [4, Lemma 5.7], in the energy normalization. We include the short proof because the main argument relies on the exact branch description and on the positivity of the weights.

Lemma 2.2. *Assume $0 < a, b \leq c$.*

(a) *If $a + b \leq c$, then $E = c/2$.*

(b) *If $a + b > c$, then $D \geq 3ab > 0$, the point x^* has three positive coordinates, $F > c/2$, and $E = F$.*

Proof. The second and third coordinates of $D\,x^*$ are $b(a + c - b) \geq ab > 0$ and $a(b + c - a) \geq ab > 0$. Suppose x^* lies in the open simplex. Then its second coordinate is positive, which forces $D > 0$; positivity of the first coordinate then forces $a + b > c$. Hence, if $a + b \leq c$, Lemma 2.1 gives $E = \max\{a, b, c\}/2 = c/2$. If instead $a + b > c$, then $0 < a + b - c \leq \min\{a, b\} \leq \sqrt{ab}$. So $D = 4ab - (a + b - c)^2 \geq 3ab > 0$, all coordinates of x^* are positive, and $x^* \in \Delta$ gives $E \geq F$. The identity

$$F - \frac{c}{2} = \frac{c\,(a + b - c)^2}{2D} \tag{5}$$

shows $F > c/2$. Therefore the alternative $E = c/2$ of Lemma 2.1 is impossible, and $E = F$. □

3 Proof of the main theorem

3.1 Chords and their ordering

Fix $\varphi \in (\pi/2, 2\pi/3]$ and $\psi \in [0, \psi_{\mathrm{e}}]$. Put

$$\theta_1 = \frac{\varphi - \psi}{2}, \qquad \theta_2 = \frac{\varphi + \psi}{2}, \qquad p = \sin^2\theta_1, \quad q = \sin^2\theta_2, \quad s = \sin^2\varphi, \quad k = p + q - s.$$

For points on the unit circle, $|e^{i\alpha} - e^{i\beta}| = 2|\sin\frac{\alpha-\beta}{2}|$. Hence

$$|A - B|^2 = 4p, \qquad |A - C|^2 = 4q, \qquad |B - C|^2 = 4s. \tag{6}$$

Geometrically, θ_1, θ_2 and $\pi - \varphi$ are the angles of the triangle ABC at C, B and A, and $k = p + q - s$ is the law of cosines written with the angle at A.

Lemma 3.1. *With the notation above:*

(a) $k = \cos\varphi\,(\cos\varphi - \cos\psi) = -2\cos\varphi\,\sin\theta_1 \sin\theta_2$, *and* $k^2 = 4\cos^2\varphi\, pq$;

(b) $0 < k \le p \le q \le s$, *and* $q = s$ *if and only if* $\psi = \psi_{\mathrm{e}}$;

(c) $|A_{\mathrm{e}} - B|^2 = 16\, s\cos^2\varphi \le 4s$ *and* $|A_{\mathrm{e}} - C| = |B - C|$.

Proof. (a) Since $\sin^2 x = (1 - \cos 2x)/2$, we have $p + q = 1 - \frac{1}{2}(\cos(\varphi - \psi) + \cos(\varphi + \psi)) = 1 - \cos\varphi\cos\psi$, and $s = 1 - \cos^2\varphi$. This gives the first formula for k. Also $2\sin\theta_1 \sin\theta_2 = \cos(\theta_1 - \theta_2) - \cos(\theta_1 + \theta_2) = \cos\psi - \cos\varphi$, which gives the second formula. Squaring it gives $k^2 = 4\cos^2\varphi\, pq$.

(b) From $\psi \le 2\pi - 3\varphi$ we get $\theta_1 \ge (\varphi - 2\pi + 3\varphi)/2 = 2\varphi - \pi > 0$ and $\theta_2 \le \pi - \varphi < \pi/2$. Also $\theta_1 \le \theta_2$ because $\psi \ge 0$. Thus $0 < \theta_1 \le \theta_2 \le \pi - \varphi < \pi/2$. Since sin is positive and strictly increasing on $(0, \pi/2]$,

$$0 < \sin\theta_1 \le \sin\theta_2 \le \sin(\pi - \varphi) = \sin\varphi,$$

so $p \le q \le s$, with $q = s$ exactly when $\theta_2 = \pi - \varphi$, that is, when $\psi = \psi_{\mathrm{e}}$. Next, $k = p - (s - q) \le p$. Finally $\cos\varphi < 0$, and $\cos\varphi - \cos\psi < 0$ because $0 \le \psi \le 2\pi - 3\varphi < \varphi \le \pi$ (using $\varphi > \pi/2$). By (a), $k > 0$.

(c) At $\psi = \psi_{\mathrm{e}}$, $\theta_1 = 2\varphi - \pi$, so $|A_{\mathrm{e}} - B|^2 = 4\sin^2(2\varphi) = 16\sin^2\varphi\cos^2\varphi$. Since $\cos\varphi \in [-\frac{1}{2}, 0)$, we have $4\cos^2\varphi \le 1$. Also $\theta_2 = \pi - \varphi$ there, so $|A_{\mathrm{e}} - C|^2 = 4\sin^2\varphi = |B - C|^2$. □

3.2 The majorization inequality

Lemma 3.2. *Let* $0 < k \le p \le q \le s$ *with* $p + q = k + s$, *and let* $t > 0$.

(a) *If* $t \ge 1$, *then* $p^t + q^t \le k^t + s^t$. *If* $t > 1$, *equality holds if and only if* $q = s$.

(b) *If* $0 < t \le 1$, *then* $p^t + q^t \ge k^t + s^t$. *If* $t < 1$, *equality holds if and only if* $q = s$.

Proof. If $k = s$ then $p = q = s = k$ and there is nothing to prove; so let $k < s$. Put $\lambda = (s - p)/(s - k) \in [0, 1]$. Then $p = \lambda k + (1 - \lambda)s$, and, because $p + q = k + s$, also $q = (1 - \lambda)k + \lambda s$. Let $f(u) = u^t$ on $(0, \infty)$. For $t \ge 1$, f is convex, and

$$f(p) + f(q) \le \lambda f(k) + (1 - \lambda)f(s) + (1 - \lambda)f(k) + \lambda f(s) = f(k) + f(s).$$

For $t > 1$, f is strictly convex. Equality then forces p and q to be endpoints of $[k, s]$, so $\lambda \in \{0, 1\}$. If $\lambda = 0$ then $p = s$, and $p \le q \le s$ gives $q = s$. If $\lambda = 1$ then $p = k$ and $q = s$. Conversely, $q = s$ gives $p = k$ and equality. For $0 < t \le 1$, f is concave and the same argument reverses every inequality. □

3.3 Comparison with the endpoint

Let $t = r/2$. By (6), the kernel values in (1) are

$$a = |A - B|^r = 2^r p^t, \qquad b = |A - C|^r = 2^r q^t, \qquad c = |B - C|^r = 2^r s^t,$$

and by Lemma 3.1(b), $0 < a \le b \le c$. Thus Lemma 2.2 applies, with c the largest value. Note that c does not depend on ψ. At the endpoint, by Lemma 3.1(c),

$$b(\psi_{\mathrm{e}}) = c, \qquad e := a(\psi_{\mathrm{e}}) = |A_{\mathrm{e}} - B|^r = 2^r(4s\cos^2\varphi)^t \le c.$$

Since $e + c > c$, Lemma 2.2(b) gives

$$E_r(\psi_{\mathrm{e}}; \varphi) = F_{\mathrm{e}} := \frac{2c \cdot c \cdot e}{4ce - e^2} = \frac{2c^2}{4c - e}, \tag{7}$$

with the masses stated in Theorem 1.2(iii). These are the coordinates of (4) at $(a, b, c) = (e, c, c)$, positive because $e \le c$. Substituting $c = 2^r \sin^r \varphi$ and $e = 2^r \cdot 2^r \sin^r \varphi \, |\cos \varphi|^r$ into (7) gives the closed form in (2).

For general $a, b > 0$ with $D > 0$, direct expansion gives

$$F_{\mathrm{e}} - \frac{2abc}{D} = \frac{2c\,[e\,ab - c\,(a+b-c)^2]}{(4c-e)\,D}, \tag{8}$$

because $cD - ab(4c-e) = e\,ab - c(a+b-c)^2$ by (3). On our family, Lemma 3.1(a) gives $4\cos^2\varphi\, pq = k^2$, so

$$e\,ab = 2^{3r}(4s\cos^2\varphi)^t p^t q^t = 2^{3r} s^t\,(4\cos^2\varphi\, pq)^t = 2^{3r} s^t k^{2t}, \qquad c\,(a+b-c)^2 = 2^{3r} s^t\, m^2,$$

where $m := p^t + q^t - s^t$. Hence

$$e\,ab - c\,(a+b-c)^2 = 2^{3r} s^t\,(k^t - m)(k^t + m). \tag{9}$$

The relation $a + b > c$ is equivalent to $m > 0$.

Proof of Theorem 1.2. Let $r \ge 2$, so $t \ge 1$. By Lemma 3.1(b), $0 < k \le p \le q \le s$ and $p + q = k + s$. Lemma 3.2(a) therefore gives $m \le k^t$, with strict inequality when $t > 1$ and $\psi < \psi_{\mathrm{e}}$. By Lemma 2.2 there are two cases.

Two-point branch ($m \le 0$, i.e. $a + b \le c$). Here $E_r(\psi;\varphi) = c/2$. By (7), $F_{\mathrm{e}} - c/2 = ce/(2(4c-e)) > 0$. So $E_r(\psi;\varphi) < E_r(\psi_{\mathrm{e}};\varphi)$.

Three-point branch ($m > 0$). Here $E_r(\psi;\varphi) = 2abc/D$ with $D > 0$. In (8), $4c - e \ge 3c > 0$ and $D > 0$, and by (9) the bracket has the sign of $k^t - m \ge 0$, because $k^t + m > 0$. Hence $E_r(\psi;\varphi) \le E_r(\psi_{\mathrm{e}};\varphi)$, strictly when $k^t > m$.

This proves (2) and (iii). For (ii), let $r > 2$ and $\psi < \psi_{\mathrm{e}}$. Then $q < s$, so $m < k^t$ by Lemma 3.2(a), and both cases are strict.

For (i), let $r = 2$, so $t = 1$ and $m = p + q - s = k > 0$. Every admissible ψ is then on the three-point branch, and the bracket (9) vanishes. So $E_2(\psi;\varphi) = F_{\mathrm{e}}$, and by (2), $F_{\mathrm{e}} = 2\sin^2\varphi/(1 - \cos^2\varphi) = 2$. □

Remark 3.3 (Where the two branches occur). For $r > 2$ both branches can occur. As ψ increases, the switch quantity $a + b - c = 2^r m$ need not be monotone [4, Section 5.4.1]. The proof above never needs to locate the switch: the two-point value is always below F_{e} by the explicit margin $ce/(2(4c-e))$, and on the three-point branch the comparison is exactly Lemma 3.2. In the original variables, the three-point branch is the set where

$$\sin^r \tfrac{\varphi-\psi}{2} + \sin^r \tfrac{\varphi+\psi}{2} > \sin^r \varphi.$$

Proposition 3.4 (The reversed regime $0 < r < 2$)**.** *Let $0 < r < 2$, $\pi/2 < \varphi \le 2\pi/3$ and $0 \le \psi \le \psi_{\mathrm{e}}$. Then every admissible ψ lies on the three-point branch, and $E_r(\psi;\varphi) \ge E_r(\psi_{\mathrm{e}};\varphi)$, with strict inequality for $\psi < \psi_{\mathrm{e}}$. So for these exponents the isosceles endpoint is the unique* minimizer *on the arc.*

Proof. Now $0 < t < 1$. Since $u \mapsto u^t$ is strictly subadditive on $(0, \infty)$, we get $p^t + q^t > (p+q)^t = (s+k)^t > s^t$. So $m > 0$ and $a + b > c$. Lemmas 3.1 and 2.2 hold for every $r > 0$; their proofs did not use $r \ge 2$. So $E_r(\psi;\varphi) = 2abc/D$ and $E_r(\psi_{\mathrm{e}};\varphi) = F_{\mathrm{e}}$. Lemma 3.2(b) gives $m \ge k^t$, strictly when $\psi < \psi_{\mathrm{e}}$. By (8) and (9), $F_{\mathrm{e}} \le 2abc/D$, strictly when $\psi < \psi_{\mathrm{e}}$. □

4 Arbitrary inscribed triangles and the odd-polygon conjecture

For $\pi/3 \le \gamma < \pi/2$ and $r > 0$ put

$$G_r(\gamma) = \frac{2^{r-1}\sin^r\gamma}{1 - 2^{r-2}\cos^r\gamma}. \tag{10}$$

The denominator is at least $3/4$, because $2\cos\gamma \le 1$. By Theorem 1.2, $G_r(\pi - \varphi) = E_r(\psi_{\mathrm{e}}; \varphi)$ is the energy of the isosceles inscribed triangle with angles $\gamma, \gamma, \pi - 2\gamma$.

Corollary 4.1 (Triangle form)**.** *Let T be the vertex set of a triangle inscribed in the unit circle with angles $\alpha \le \beta \le \gamma$, and suppose $\gamma < \pi/2$. If $r \ge 2$, then $U_r(T) \le G_r(\gamma)$. If $r > 2$, equality holds only when $\beta = \gamma$.*

Proof. Necessarily $\gamma \ge \pi/3$. Put $\varphi = \pi - \gamma \in (\pi/2, 2\pi/3]$ and $\psi = \beta - \alpha \ge 0$. Then $\theta_1 = (\varphi - \psi)/2 = \alpha$ and $\theta_2 = (\varphi + \psi)/2 = \beta$. The condition $\beta \le \gamma$ is equivalent to $\psi \le 2\pi - 3\varphi$. The points $A = e^{i\psi}$, $B = e^{i\varphi}$, $C = e^{-i\varphi}$ form an inscribed triangle with the angles $\pi - \varphi = \gamma$ at A, $\theta_2 = \beta$ at B and $\theta_1 = \alpha$ at C. By the chord formula (6) it has the same side lengths as T, so it is congruent to T. Since U_r is invariant under isometries, $U_r(T) = E_r(\psi; \varphi) \le G_r(\gamma)$ by Theorem 1.2. For $r > 2$, equality requires $\psi = \psi_{\mathrm{e}}$, that is, $\beta = \gamma$. □

Lemma 4.2 (cf. Fan [4, Lemma 5.8])**.** *On $[\pi/3, \pi/2)$ the function G_2 is identically 2, and G_r is strictly increasing for every $r > 2$.*

Proof. Write $S = \sin\gamma$, $C = \cos\gamma$ and $\Theta = 1 - 2^{r-2}C^r > 0$. Then

$$\frac{d}{d\gamma}\log G_r = \frac{rC}{S} - \frac{2^{r-2}rC^{r-1}S}{\Theta} = \frac{rC}{S\Theta}\Big(1 - 2^{r-2}C^r - 2^{r-2}C^{r-2}S^2\Big) = \frac{rC}{S\Theta}\big(1 - (2C)^{r-2}\big).$$

For $\gamma \in (\pi/3, \pi/2)$ we have $0 < 2C < 1$, so the derivative is positive when $r > 2$ and zero when $r = 2$; also $G_2(\pi/3) = 2$. □

We now turn to Fan's odd-polygon conjecture. Fan states it with the equilibrium-support theorem of Björck, which we use as an imported result and do not reprove.

Theorem 4.3 (Björck [1, Theorem 12], as stated in [2, Section 11] and [4, Section 5.1])**.** *Let $r > 2$ and let $K \subset \mathbb{R}^n$ be compact and contain at least two points. Then K contains a set E of at most $n + 1$ points with $U_r(E) = U_r(K)$.*

Corollary 4.4 (Fan's Conjecture 5.6)**.** *Let $r > 2$, let $N \ge 3$ be odd, let P_N be the vertex set of a regular N-gon inscribed in the unit circle, and let $\theta_N = (N - 1)\pi/(2N)$. Assume Theorem 4.3. Then*

$$U_r(P_N) = G_r(\theta_N) = \frac{2^{r-1}\sin^r\theta_N}{1 - 2^{r-2}\cos^r\theta_N}.$$

An equilibrium measure is supported on the three points $1, e^{2\theta_N i}, e^{-2\theta_N i}$ (up to rotation). With $a = 2\sin(\pi/N)$ and $b = 2\sin\theta_N$, its masses are

$$\frac{2b^r - a^r}{4b^r - a^r}, \qquad \frac{b^r}{4b^r - a^r}, \qquad \frac{b^r}{4b^r - a^r}.$$

Proof. By Theorem 4.3 with $n = 2$, $U_r(P_N)$ is the maximum of $U_r(T)$ over subsets $T \subset P_N$ with at most three points. Every such subset lies in a three-point subset, and U_r is monotone under inclusion. So it suffices to consider vertex triangles. An inscribed angle subtending j sides of the N-gon equals $j\pi/N$. So a vertex triangle has angles $\alpha \le \beta \le \gamma$ equal to $i\pi/N \le j\pi/N \le k\pi/N$ with $i + j + k = N$ and $i \ge 1$. Since N is odd, $\gamma \ne \pi/2$.

Obtuse triangles ($k \geq (N+1)/2$). Put $P = \sin^2\alpha$, $Q = \sin^2\beta$, $S = \sin^2\gamma$. The computation in Lemma 3.1(a) uses no restriction on φ or ψ. With $\varphi = \pi - \gamma$ and $\psi = \beta - \alpha$ it gives $P + Q - S = 2\cos\gamma\sin\alpha\sin\beta < 0$. Moreover $\alpha, \beta < \pi - \gamma$, so S exceeds P and Q, and the side opposite γ is the longest. Since $u \mapsto u^t$ is superadditive for $t = r/2 \geq 1$, we get $P^t + Q^t \leq (P+Q)^t < S^t$. So Lemma 2.2(a) applies, and the energy is $\frac{1}{2}(2\sin\gamma)^r$. Because sin decreases on $[\pi/2, \pi]$, we have $\sin\gamma \leq \sin\frac{(N+1)\pi}{2N} = \cos\frac{\pi}{2N} = \sin\theta_N$. Therefore $U_r(T) \leq \frac{1}{2}(2\sin\theta_N)^r < G_r(\theta_N)$, the last step by (5) applied to the isosceles triangle.

Acute triangles ($k \leq (N-1)/2$). Here $\pi/3 \leq \gamma \leq \theta_N$, so Corollary 4.1 and Lemma 4.2 give $U_r(T) \leq G_r(\gamma) \leq G_r(\theta_N)$.

Attainment. The triangle with $(i, j, k) = (1, \frac{N-1}{2}, \frac{N-1}{2})$ is the set $\{1, e^{2\theta_N i}, e^{-2\theta_N i}\}$, since $2\theta_N = 2\pi \cdot \frac{N-1}{2}/N$. It is isosceles with apex 1, legs b and base $|e^{2\theta_N i} - e^{-2\theta_N i}| = 2\sin(2\theta_N) = 2\sin(\pi/N) = a$. By Theorem 1.2(iii), with $c = b^r$ and $e = a^r$, its energy is $2b^{2r}/(4b^r - a^r) = G_r(\theta_N)$. It is attained by the stated masses; the apex receives $(2b^r - a^r)/(4b^r - a^r)$. This probability measure on P_N attains $U_r(P_N)$, so it is an equilibrium measure. □

Remark 4.5. Corollary 4.4 gives the energy and *an* equilibrium measure. We make no uniqueness claim, either up to rotation or otherwise. Its only external input is Theorem 4.3. We take that theorem from the cited literature and have not checked Björck's original proof. For $N = 3$ the statement is the equilateral case: $G_r(\pi/3) = \frac{2}{3}\,3^{r/2}$, with equal masses.

5 Computational cross-checks

The proofs above are complete without computation. As independent safeguards we ran two kinds of checks; both scripts are in the reproducibility package.

- *Exact symbolic verification* (SymPy). This verifies 26 identities by exact simplification to zero: the critical-point formula (4), its value, $\det M$, $\mathbf{1}^{\mathsf{T}}M^{-1}\mathbf{1}$, the identity (5), the endpoint masses and value (7), the key identity (8), every trigonometric identity of Lemma 3.1 together with the chord formulas (6), the convex-combination representation in Lemma 3.2, and the closed form (10).
- *Floating-point regression tests*, which are not part of the proof. (A) For 253 parameter triples, including $r = 2$, $\varphi = 2\pi/3$, $\psi = 0$ and $\psi = \psi_{\mathrm{e}}$, we maximized Q over the simplex by grid search with zooming refinement, using no formula. The result agreed with Lemma 2.2 to relative error below $5 \cdot 10^{-16}$. (B) We evaluated $E_r(\psi;\varphi)$ at 627,120 grid points with $r \in [2, 200]$. The largest relative excess over the endpoint value was $6 \cdot 10^{-15}$, which is rounding. For $r = 2$, E equals 2 to within $9 \cdot 10^{-16}$. We also computed the gap in 600-digit arithmetic for $r \geq 2.1$ and $\psi \leq 0.9\,\psi_{\mathrm{e}}$. It was positive throughout, as low as about 10^{-473}. On the two-point branch the gap equals $e/(4c - e) \approx (2|\cos\varphi|)^r/4$, which is astronomically small when r is large and φ is close to $\pi/2$; this is why the strictness check needs high precision. (C) The endpoint comparison also held using only brute-force maxima. (D) For odd $N \leq 31$ and $r \in \{2.5, 3, 6, 15\}$, the maximum over all vertex triples of P_N equaled $G_r(\theta_N)$ to relative error $4 \cdot 10^{-15}$. Replicator iterations $x_i \mapsto x_i(Mx)_i/x^{\mathsf{T}}Mx$ over all measures on P_N, from random starts, never exceeded it by more than rounding error (relative $3 \cdot 10^{-15}$). (E) For $0 < r < 2$, on 3960 comparisons, the three-point branch was always active and the endpoint was never larger than $E_r(\psi;\varphi)$.

6 Related work, novelty and scope

Three-point capacities with negative exponents were computed by Clark and Laugesen [2, Theorem 8]. The capacity-ratio program and the support theorem of Björck [1] are discussed

there and in [3, 4]. Fan [4] proved the symmetric step [4, Lemma 5.8], formulated Conjecture 1.1, and noted that it implies the odd-polygon Conjecture 5.6. Fan's attempt studied the sign changes of $a + b - c$ along the arc. We checked the following sources on 28 October 2026: arXiv:2609.11186 (only v1, 10 September 2026), the associated code repository of Fan (README: the odd-polygon conjecture "unproved"), arXiv full-text and listing searches for "Riesz capacity", and general web searches. We found no proof of Conjecture 1.1 or of Conjecture 5.6. To the best of our knowledge, Theorem 1.2, Proposition 3.4 and Corollary 4.1 are new. Lemmas 2.2 and 4.2 re-prove known results of Clark–Laugesen and Fan. Corollary 4.4 is conditional only on the published Theorem 4.3. The paper proves exactly the statements labelled as theorems, propositions, lemmas and corollaries. Monotonicity of $E_r(\cdot\,;\varphi)$ in ψ, uniqueness of equilibrium measures on P_N, and the capacity-ratio questions of [4] are not addressed.

Authorship and verification. Claude Opus 5.5, an AI system developed by Anthropic, found and wrote the proofs, the verification scripts and this text. No external peer review or proof-assistant formalization is claimed. Correctness rests on the proofs as written. The computational checks of Section 5 are supplementary.

References

[1] G. Björck, Distributions of positive mass, which maximize a certain generalized energy integral, *Ark. Mat.* **3** (1956), 255–269.

[2] C. Clark and R. S. Laugesen, Maximizing Riesz capacity ratios: conjectures and theorems, *SIAM J. Math. Anal.* **57** (2025), no. 5, 5683–5717; arXiv:2410.14809v1.

[3] C. Clark and R. S. Laugesen, Riesz capacity: monotonicity, continuity, diameter and volume, *Anal. Math. Phys.* **15** (2025), Paper 4.

[4] Q. Fan, Riesz capacity ratios with negative exponents, arXiv:2609.11186v1 (10 September 2026), Conjectures 5.6 and 5.9, Lemmas 5.7 and 5.8. `https://arxiv.org/abs/2609.11186`

A Reproducibility

The package contains this source (`paper/main.tex`), the symbolic verifier (`src/verify_symbolic.py`, requiring SymPy), the floating-point regression script (`src/verify_numeric.py`, requiring NumPy and mpmath), the build script (`src/build_paper.py`, requiring pdfLaTeX), and the generated reports in `certificates/`. Paths are relative to the package root. The scripts locate their files relative to their own location, so they run from any working directory.